[20241478 최주영] Assignment6 - Report

생성자	ichbinju0

1. Introduction
   a. Background

      The ‘1,000won Breakfast’ program is a government-subsidized meal initiative that provides affordable morning meals to university students, funded by a combination of government support (2,000 KRW), student payment (1,000 KRW), and a university subsidy. Since the university bears the remaining cost, minimizing institutional expenditure while maintaining meal quality is a key operational objective. This problem can be viewed as a production planning problem in the context of Operations Management, where the daily menu composition represents a repeated production decision that must balance cost efficiency with output quality.

   b. Motivation

      This program is particularly well-suited for MIP-based optimization for two reasons.

      - The food items provided are commercially available products, mostly minimally processed whole foods, with nutritional information and retail prices clearly labeled, making data collection straightforward.
      - Given the small scale of approximately 30 meals per session at UNIST, procurement sources are relatively predictable compared to large-scale institutional food contracts, where unit costs are difficult to estimate.

   c. Problem Statement

      This study applies Mixed Integer Programming (MIP) to formulate the breakfast menu composition as a cost minimization problem. The objective is to find the optimal combination of food items that minimizes total cost per serving while satisfying calorie and nutrient ratio requirements, within a budget range of 4,000 to 6,000 KRW, reflecting the total funding available across all three sources.

2. Problem Definition
   a. Input Variables

      The input variables of this model are the nutritional coefficients and unit prices of each food item, collected from product nutrition labels and retail prices. These include calories (kcal), carbohydrates (g), and protein (g) per unit, as well as price (KRW) per unit, for all 13 candidate food items.

Category	Food Name	Varaiable Type	Unit	Carlories(kcal/unit)	Carbohydrates(g/unit)	Protein(g/unit)	Price(KRW/unit)
Main	Shefood Tuna Mayo Triangle Kimbap	Binary	1 item (110g)	205	33	5	1660
Main	Samlip Glutinous Milk Bread	Continuous	per g	2.66	0.47	0.06	6.9
Main	Red Bean Bread	Binary	1 item (85g)	250	42	7	1232.5
Main	Sweet Potato	Continuous	per g	1.28	0.3036	0.0172	4.97
Drink	Maeil Soy milk	Binary	1 pack (190ml)	95	4	9	657.4
Drink	Orange Juice	Binary	1 pack (195ml)	44	11	0	302.1
Drink	Seoul Milk	Binary	1 pack (200ml)	135	10	6	536
Protein	Bodeure Soft Boiled Egg	Integer	1 item (50g)	70	0	6	530
Protein	Chicken Breast Bar	Binary	1 item (80g)	130	3	12	1304
Protein	Protein Bar	Binary	1 item (50g)	260	18	14	1049
Side	Seoul Milk Pure Plain Yogurt	Continuous	per ml	0.7	0.05	0.03	3.8
Side	Market O Nature Granola	Continuous	per g	4.83	0.72	0.06	17.53
Side	Banana	Integer	1 item (100g)	105	26.95	1.29	349

   b. Decision Variables

      The model includes 13 primary decision variables representing the serving amount of each food item, along with 4 auxiliary binary variables (y2, y4, y11, y12) introduced for Big-M constraints to link continuous variables to selection indicators.

Variable	Food Item	Type	Range
x1	Shefood Tuna Mayo Triangle Kimbap	Binary	{0,1} item
x2	Samlip Glutinous Milk Bread	Continuous	0~150g
x3	Red Bean Bread	Binary	{0,1} item
x4	Sweet Potato	Continuous	0~200g
x5	Maeil Soy milk	Binary	{0,1} pack
x6	Orange Juice	Binary	{0,1} pack
x7	Seoul Milk	Binary	{0,1} pack
x8	Bodeure Soft Boiled Egg	Integer	0~2 items
x9	Chicken Breast Bar	Binary	{0,1} item
x10	Protein Bar	Binary	{0,1} item
x11	Seoul Milk Pure Plain Yogurt	Continuous	0~200ml
x12	Market O Nature Granola	Continuous	0~50g
x13	Banana	Integer	0~2 items

c. Objective Function

The objective is to minimize total meal cost per serving while penalizing deviation from the calorie target of 450 kcal.

$$\text{Minimize} \sum_{i=1}^{13} p_i x_i + w(d^+ + d^-)$$

where d^+ and d^- are over and under deviation from the calorie target respectively.

d. Output Variables

The output of this model is the optimal menu composition, the selected food items and their serving amounts, along with the resulting total cost, calorie count, and nutrient ratios.

e. Constraints

#	Constraint	Expression	Reason
1	Calorie Range	300≤kcal≤600	Reflects an appropriate calorie intake for a light breakfast; lower bound prevents a nutritionally insufficient meal, upper bound prevents over-eating in the morning
2	Calorie target deviation	cal − d^+ + d^- = 450	Penalizes deviation from the target of 450 kcal (midpoint of the range) to encourage balanced meal planning
3	Carbohydrates ratio	50% ≤ carb×4/kcal ≤ 65%	Based on the Korean Dietary Reference Intakes 2025 (Ministry of Health and Welfare, ages 19–29)
4	Protein ratio	10% ≤ pro×4/kcal ≤ 20%	Based on the Korean Dietary Reference Intakes 2025 (Ministry of Health and Welfare, ages 19–29)
5	Budget	4,000 ≤ cost ≤ 6,000 KRW	Reflects the total funding available per meal (2,000 KRW government + 1,000 KRW student + university subsidy); lower bound ensures minimum meal quality
6	Exactly 1 main item	x1 + y2 + x3 + y4 = 1	One main dish per meal is a basic composition principle; prevents redundant carbohydrate sources
7	Exactly 1 drink	x5 + x6 + x7 = 1	One drink per meal is sufficient; prevents unnecessary cost and calorie overlap
8	At least 1 protein	x8 + x9 + x10 ≥ 1	A protein source is essential for a nutritionally complete breakfast
9	Big-M: Milk Bread	x2 ≤ 150·y2	Links continuous amount to binary selection; 150g set as a realistic single serving upper limit

#	Constraint	Expression	Reason
10	Big-M: Sweet Potato	$x4 \leq 200 \cdot y4$	Links continuous amount to binary selection; 200g corresponds to approximately 2 medium sweet potatoes
11	Big-M: Yogurt	$x11 \leq 200 \cdot y11$	Links continuous amount to binary selection; 200ml is a reasonable single serving for plain yogurt
12	Big-M: Granola	$x12 \leq 50 \cdot y12$	Links continuous amount to binary selection; 50g is a standard single serving for granola
13	Yogurt–Granola set	$y11 = y12$	Yogurt alone lacks texture and granola alone is too dry; serving them together reflects common consumption practice
14	Granola minimum	$x12 \geq 10 \cdot y12$	Ensures a meaningful serving of granola (at least 10g) if selected; prevents a negligible amount being chosen solely to satisfy the constraint

3. Data Collection

   Data for this assignment were collected from two primary sources: product nutrition labels and prices. All 13 food items were selected based on menus actually served in the 1,000 Won Breakfast program, as I observed through direct participation.

   - Nutritional ratio standards: Dietary Reference Intakes for Koreans 2025, Ministry of Health and Welfare
   - Nutritional Data & Price Data

     Nutritional information was primarily collected from product descriptions on Coupang, where most items display nutrition labels in their product detail pages. For items whose nutritional information was not explicitly listed on Coupang, data were supplemented using FatSecret. All values are based on the Serving Unit defined for each item in the model. Unit prices were collected from Coupang, reflecting retail consumer prices at the time of data collection. For items sold in bulk, the per-unit price was calculated by dividing the total package price by the number of units.

     1. Shefood Tuna Mayo Triangle Kimbap
        - Source: https://www.coupang.com/vp/products/8687721625?itemId=25222599427&vendorItemId=92218940049&pickType=COU_PICK&q=참치마요+삼각김밥&searchId=c701493d9156901&sourceType=search&itemsCount=60&searchRank=1&rank=1&traceId=moi
        - Nutrition Facts

영양정보	총 내용량 330 g / 110 g 당 205 kcal	
나트륨 500 mg 25 %	탄수화물 33 g 10 %	당류 1 g 1 %
지방 6.0 g 11 %	트랜스지방 0 g	포화지방 1.7 g 11 %
콜레스테롤 5 mg 미만 2 %	단백질 5 g 9 %	

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

        - Price: 1,660 KRW per item
     2. Samlip Glutinous Milk Bread
        - Source: https://www.coupang.com/vp/products/5453525180?itemId=8330922846&vendorItemId=75618767406&q=통식빵&searchId=520ce1c09554062&sourceType=search&itemsCount=60&searchRank=1&rank=1&traceId=mo
        - Nutrition Facts

영양정보	총 내용량 840 g 100 g당 266 kcal
100 g 당	1일 영양성분 기준치에 대한 비율
나트륨 480 mg	24 %
탄수화물 47 g	15 %
당류 9 g	9 %
지방 6 g	11 %
트랜스지방 0 g	
포화지방 2.4 g	16 %
콜레스테롤 5 mg	2 %
단백질 6 g	11 %

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

- Price: 6.9 KRW per g

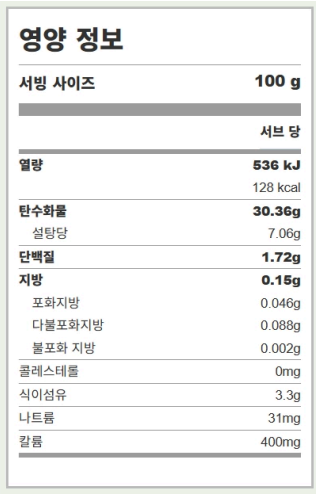
3. Red Bean Bread
   - Source: https://www.coupang.com/vp/products/7119010350?itemId=17819482398&vendorItemId=84983394093&pickType=COU_PICK&q=단팥빵&searchId=4fc5d9e19764801&sourceType=search&itemsCount=60&searchRank=1&rank=1&traceId=moit
   - Nutrition Facts

영양정보	총 내용량 255 g (85 g x 3개) 1개 (85 g)당 250 kcal
1개당	1일 영양성분 기준치에 대한 비율
나트륨 190 mg	10 %
탄수화물 42 g	13 %
당류 18 g	18 %
지방 6 g	11 %
트랜스지방 0 g	
포화지방 2.8 g	19 %
콜레스테롤 0 mg	0 %
단백질 7 g	13 %

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

   - Price: 1,232.5 KRW per item

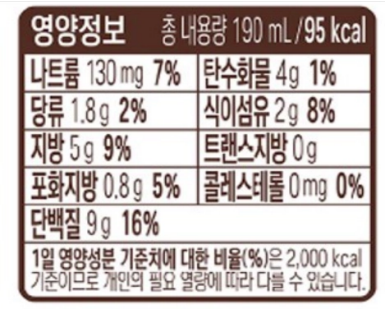
4. Sweet Potato
   - Source: https://www.coupang.com/vp/products/9476338691?itemId=28210253905&vendorItemId=95156598859&sourceType=srp_product_ads&clickEventId=8d002b431c-11f1-88af-e0177d3aa5f9&korePlacement=15&koreSubPlacement=1&clickEventId=8d002be0-431c-11f1-88af-e0177d3aa5f9&korePlacement=15&koreSubPlacement=1&traceId=moitmmdi
   - Nutrition Facts



영양 정보	
서빙 사이즈	100 g
	서브 당
열량	536 kJ
	128 kcal
탄수화물	30.36g
설탕당	7.06g
단백질	1.72g
지방	0.15g
포화지방	0.046g
다불포화지방	0.088g
불포화 지방	0.002g
콜레스테롤	0mg
식이섬유	3.3g
나트륨	31mg
칼륨	400mg

- Price: 4.97 KRW per g

5. Maeil Soy milk
   - Source: https://www.coupang.com/vp/products/7757115239?itemId=20909737628&vendorItemId=3038558667&q=매일두유&searchId=fe72fb8d6873054&sourceType=search&itemsCount=60&searchRank=0&rank=0&traceId=mc
   - Nutrition Facts



영양정보	총 내용량 190 mL / 95 kcal
나트륨 130 mg 7%	탄수화물 4 g 1%
당류 1.8 g 2%	식이섬유 2 g 8%
지방 5 g 9%	트랜스지방 0 g
포화지방 0.8 g 5%	콜레스테롤 0 mg 0%
단백질 9 g 16%	
1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.	

- Price: 657.4 KRW per pack

6. Orange Juice
   - Source: https://www.coupang.com/vp/products/7384640708?itemId=19330958171&vendorItemId=93896552692&q=오렌지주스&searchId=6711df019846012&sourceType=search&itemsCount=57&searchRank=9&rank=9&traceId=moit
   - Nutrition Facts

영양정보	총 내용량 195 ml 44 kcal		
나트륨 30 mg 2 %	탄수화물 11 g 3 %	당류 10 g 10 %	
지방 0 g 0 %	트랜스지방 0 g	포화지방 0 g 0 %	
콜레스테롤 0mg 0%	단백질 0g 0%	칼슘 51mg 7%	비타민C 6mg 6%
1일 영양성분 기준치에 대한 비율(%)은 2,000kcal 기준이므로 개인의 필요열량에 따라 다를 수 있습니다.			

- Price: 302.1 KRW per pack

7. Seoul Milk
   - Source: https://www.coupang.com/vp/products/4975075206?itemId=20185206891&vendorItemId=92283920213&q=우유&searchId=24ce10819235057&sourceType=search&itemsCount=60&searchRank=6&rank=6&traceId=mc

- Nutrition Facts

영양정보			총 내용량 200 mL/ 135 kcal		
나트륨	100 mg	5%	콜레스테롤	25 mg	8%
탄수화물	10 g	3%	단백질	6 g	11%
당류	10 g	10%	칼슘	200 mg	29%
지방	8 g	15%			
트랜스지방	0.5 g 미만				
포화지방	5 g	33%			

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

- Price: 536 KRW per pack

8. Bodeure Soft Boiled Egg
   - Source: https://www.coupang.com/vp/products/8148606556?itemId=23178841230&vendorItemId=90211608366&q=아침식사대용+간편식&searchId=03ac18a29365644&sourceType=search&itemsCount=60&searchRank=2&rank=2&traceId=m
   - Nutrition Facts

분류	영양성분	100g 당 함량	100 적용	1일 영양성분 기준치 (%)
일반성분				
	에너지	140.00kcal	**140.00kcal**	7%
	탄수화물	0.00g	**0.00g**	0%
	당류	0.00g	**0.00g**	0%
	지방	9.80g	**9.80g**	18.15%
	단백질	12.00g	**12.00g**	21.82%

   - Price: 530 KRW per unit

9. Chicken Breast Bar
   - Source: https://www.coupang.com/vp/products/8279527060?itemId=23927029419&vendorItemId=90949201949&sourceType=srp_product_ads&clickEventId=a7d9c1 431b-11f1-b68e-b795c7eb3336&korePlacement=15&koreSubPlacement=3&clickEventId=a7d9c1c0-431b-11f1-b68e-b795c7eb3336&korePlacement=15&koreSubPlacement=3&traceId=moiteewp
   - Nutrition Facts

   - Price: 1,304 KRW per unit

10. Protein Bar
    - Source: https://www.coupang.com/vp/products/332471584?itemId=1062234467&vendorItemId=5541640992&q=프로틴바&searchId=86d9fb6c554332&sourceType=search&itemsCount=59&searchRank=2&rank=2&traceId=mc
    - Nutrition Facts

영양정보	총 내용량 600 g(50 g x 12개) 1개(50 g)당 **260 kcal**	
나트륨 90 mg **5 %**	탄수화물 18 g **6 %**	당류 10 g **10 %**
지방 14 g **26 %**	트랜스지방 0 g	포화지방 2.9 g **19 %**
콜레스테롤 0 mg **0 %**	단백질 14 g **25 %**	

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 열량에 따라 다를 수 있습니다.

- Price: 1,049 KRW per unit

11. Seoul Milk Pure Plain Yogurt
    - Source: https://www.coupang.com/vp/products/296352196?itemId=9541396741&vendorItemId=76826126943&q=요거트&searchId=b0d0199f7296549&sourceType=search&itemsCount=60&searchRank=2&rank=2&traceId=m
    - Nutrition Facts

영양정보	총 내용량 2.45 L 100 mL당 **70 kcal**	
100 mL당		1일 영양성분 기준치에 대한 비율
나트륨	50 mg	3%
탄수화물	5 g	2%
당류	4 g	4%
지방	4 g	7%
트랜스지방	0.1 g	
포화지방	2.5 g	17%
콜레스테롤	15 mg	5%
단백질	3 g	5%
칼슘	100 mg	14%

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

•영양정보의 당류 함량은 우유 자체 성분의 유당 함량입니다.

K-MILK 국산우유사용인증 · 안전관리인증 HACCP · 8 801115 213536

- Price: 3.8 KRW per ml

12. Market O Nature Granola
    - Source: https://www.coupang.com/vp/products/8562175754?itemId=24666769439&vendorItemId=91737670170&q=그래놀라&searchId=284dc0f810923&sourceType=search&itemsCount=60&searchRank=2&rank=2&traceId=mois
    - Nutrition Facts

영양정보	총 내용량 360 g 100 g 당 **483 kcal**
100 g 당	1일 영양성분 기준치에 대한 비율
나트륨 200 mg	**10** %
탄수화물 72 g	**22** %
당류 4.4 g	**4** %
지방 19 g	**35** %
트랜스지방 0 g	
포화지방 10 g	**67** %
콜레스테롤 5 mg 미만	**1** %
단백질 6 g	**11** %
칼슘 105 mg	**15** %
철분 1.8 mg	**15** %
아연 1.2 mg	**14** %
비타민A 105 μgRAE	**15** %
비타민C 15 mg	**15** %
비타민B_1 0.12 mg	**10** %
비타민B_2 0.15 mg	**11** %
비타민B_6 0.18 mg	**12** %
비타민E 11 mgα-TE	**100** %
나이아신 1.6 mgNE	**11** %
엽산 40 μgDFE	**10** %

1일 영양성분 기준치에 대한 비율(%)은 2,000 kcal 기준이므로 개인의 필요 열량에 따라 다를 수 있습니다.

- Price: 17.53 KRW per g

13. Banana

- Source: https://www.coupang.com/vp/products/6398359177?itemId=591060021&vendorItemId=4547871436&q=바나나&searchId=e2ed01cc256860&sourceType=search&itemsCount=60&searchRank=2&rank=2&traceId=mc
- Nutrition Facts

영양 정보

서빙 사이즈
1 개 중형 (길이 18 cm - 20 cm)

	서브 당
열량	**439 kJ**
	105 kcal
탄수화물	**26.95g**
설탕당	14.43g
단백질	**1.29g**
지방	**0.39g**
포화지방	0.132g
다불포화지방	0.086g
불포화 지방	0.038g
콜레스테롤	0mg
식이섬유	3.1g
나트륨	1mg
칼륨	422mg

- Price: 349 KRW per unit

4. Mdoel & Results

a. Optimal Solution

The MIP model was solved using the CBC solver via PuLP in Python. The optimal breakfast menu composition is presented below.

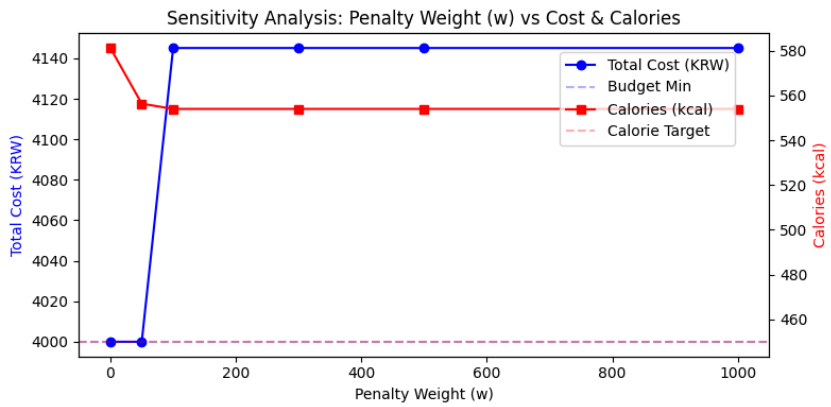
Food	Category	Serving
Shefood Tuna Mayo Triangle Kimbap	Main	1 item(s) (110g)
Orange Juice	Drink	1 pack(s) (195ml)
Bodeure Soft Boiled Egg	Protein	1 item(s) (50g)
Chicken Breast Bar	Protein	1 item(s) (80g)
Banana	Side	1 item(s) (100g)

Nutritional Summary	Result	Status
Calories (target: 450 kcal)	554.0	✅ (300~600)
Carbohydrates	53.4 %	✅ (50~65%)
Protein	17.5 %	✅ (10~20%)
Total Cost (KRW)	4145	✅ (4,000~6,000)

All constraints were satisfied in the optimal solution. The model selected one item from each requried category - main, drink and protein - while also including a side item to help meet the budget lower bound and nutritional requirements. Notably, two protein items were selected simultanously, which is permitted under the constratint of “at least one protein item”

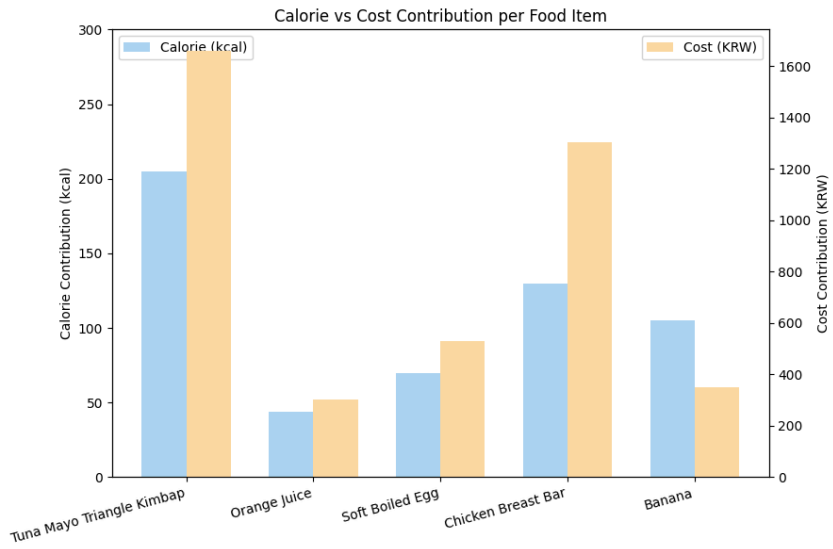
b. Sensitivity Analysis: Penalty Weight(w)

To determine the appropriate value of the penalty weight w, a sensitivity analysis was conducted by varying w from 0 to 1,000. As shown in the figure below, when w is low, the solver prioritizes cost minimization and allows calorie deviation, resulting in a total cost near the budget lower bound of 4,000 KRW while calories remain around 580 kcal. As w increases beyond 100, the solver shifts priority toward calorie accuracy, stabilizing calories at approximately 554 kcal, while cost increases to around 4,145 KRW and remains flat. Based on this analysis, w = 500 was selected, as it sufficiently penalizes calorie deviation without distorting the cost structure.



c. Calorie vs Cost contribution per Food Item

The bar chart compares the calorie and cost distribution of each selected food item. Tuna Mayo Triangle Kimbap contributes the most calories but also carries the highest cost, making it a relatively expensive calorie source. In contrast, Orange Juice and Banana provide moderate calories at low cost, improving the overall cost efficiency of the meal. Chicken Breast Bar has the second-highest cost but provides the highest protein contribution among all selected items, justifying its inclusion from a nutritional standpoint.



5. Discussion

a. Trial and Error

The final model was reached through several iterations of constraint design, each of which revealed important characteristics of the food items and the problem structure.

- The first major challenge arose when attempting to include a fat ratio constraint. Analysis of the candidate food items relvealed that all proteing category items far exceed the 30% fat ratio threshold. Since the model simultaneously requires at least one protein item, the fat constraint and the protein requirement were structurally incompatible, making the problem infeasible. The fat constraint was therefore removed, and this limitation is acknowledged as a structural characteristic of the selected food items rather than a modeling error.

- A similar issue occurred when a sodium constraint was introduced. The Tuna Mayo Triangle Kimbap alone contributes 500 mg of sodium, and combining it with Soft Boiled Egg immediately exceeds the limit. This made it impossible to select both items simultaneously, severely restricting feasible combinations. The sodium constraint was also removed for the same reason.
- The calorie range was also adjusted during development. An initial range of 500–650 kcal proved too narrow given the food item composition, as satisfying both the calorie lower bound and the nutrient ratio constraints simultaneously was difficult. The range was relaxed to 300–600 kcal to allow the solver sufficient flexibility while still reflecting a realistic breakfast portion

b. Limitations of the Model

Despite producing a feasible and cost-efficient optimal solution, the model has several limitations worth noting.

First, the prices used in this model are based on retail consumer prices from Coupang, not actual institutional procurement prices. In practice, the university likely procures food items through bulk contracts at lower unit costs. The model therefore overestimates the true cost per meal, and the optimal solution may differ under actual procurement conditions.

Second, and more fundamentally, the model finds the mathematically optimal solution but optimality in this context is defined purely by cost and nutritional constraints, with no consideration of student preferences or meal variety. If the same optimal menu were served repeatedly, students would likely lose interest over time, reducing program participation. In reality, meal planning must balance nutritional efficiency with variety and palatability to sustain engagement. A more comprehensive model could incorporate preference data collected through student surveys, or introduce a diversity constraint that prevents the same menu from being repeated within a given period. This reflects a broader limitation of mathematical programming models: they optimize what can be quantified, but real world decisions often depend equally on factors that are difficult to express as numbers.